Target SQL

- **Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:**

1. Data type of all columns in the "customers" table.

Field name	Type	Mode	Key	Collation	Default Value	Policy Tags	Description
customer_id	STRING	NULLABLE	-	-	-	-	-
customer_unique_id	STRING	NULLABLE	-	-	-	-	-
customer_zip_code_prefix	INTEGER	NULLABLE	-	-	-	-	-
customer_city	STRING	NULLABLE	-	-	-	-	-
customer_state	STRING	NULLABLE	-	-	-	-	-

Insights – customer_id is a string.
customer_unique_id is a string.
customer_zip_code_prefix is an integer type.
customer_city is a string.
customer_state is a string.

2. Get the time range between which the orders were placed.

```
SELECT MIN (order_purchase_timestamp) AS first_order,
MAX (order_purchase_timestamp) AS last_order
FROM `Target_BCS.orders`
```

JOB INFORMATION | RESULTS | CHART | JSON

Row	first_order	last_order
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

Insights – First order was placed on 2016-09-04.
Last order was placed on 2018-10-17.

3. Count the Cities & States of customers who ordered during the given period.

```
SELECT COUNT (DISTINCT customer_city) AS Total_diff_cities , COUNT(DISTINCT
customer_state) AS Total_diff_states
FROM `Target_BCS.orders`
INNER JOIN `Target_BCS.customers`
USING (customer_id)
```

JOB INFORMATION RESULTS CHART

Row	Total_diff_cities	Total_diff_states
1	4119	27

Insights – The total number of various cities from which the orders came = 4119
The total number of various states from which the orders came = 27

- **In-depth Exploration:**

1. Is there a growing trend in the no. of orders placed over the past years?

```
SELECT EXTRACT (YEAR FROM order_purchase_timestamp) AS Year,
EXTRACT (MONTH FROM order_purchase_timestamp) AS Month,
COUNT (DISTINCT order_id) AS Total_orders
FROM `Target_BCS.orders`
GROUP BY Year, Month
ORDER BY Year, Month
```

JOB INFORMATION RESULTS CHART JSON

Row	Year	Month	Total_orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700
9	2017	6	3245
10	2017	7	4026
11	2017	8	4331
12	2017	9	4285
13	2017	10	4631
14	2017	11	7544

Insights – Initially when people were not about the stores the order remained less and when people found out about the stores the orders started increasing and became constant until the festival season of 2017 arrived when a sudden jump in orders can be seen during the months of Nov, Dec & Jan. Again, the orders were constant for next few months.

Recommendations –

a) Workforce can be increased during the festive season to meet the increased demands.
b) Stacking inventories is advised throughout this period.
c) Special offers and discounts can be run for customers so that the deal becomes lucrative.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
SELECT FORMAT_DATE ('%m', order_purchase_timestamp) AS Month,
COUNT (DISTINCT order_id) AS Total_orders
FROM `Target_BCS.orders`
GROUP BY Month
ORDER BY Month
```

JOB INFORMATION | RESULTS | CHART | JSO

Row	Month	Total_orders
1	01	8069
2	02	8508
3	03	9893
4	04	9343
5	05	10573
6	06	9412
7	07	10318
8	08	10843
9	09	4305
10	10	4959
11	11	7544
12	12	5674

Insights – If the data would have been more, we could have seen that orders during the festive season are a little more than other months. In the above data only 1 festive season is covered which doesn't provide much insight but with the data in hand we can see that first 8 months have higher sales when compared to last 4. Also, sales in the middle of the year are the highest, i.e.- May to August have the highest sales. And there after the sales declined drastically, becoming half.

Recommendations –

a) Workforce can be increased from May to August the festive season to meet the increased demands.
b) Stacking inventories is advised throughout this period.
c) Special offers and discounts can be run for customers so that the deal becomes lucrative.

3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

```
SELECT
CASE
WHEN EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'
WHEN EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN
'Mornings'
WHEN EXTRACT (HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN
'Afternoon'
ELSE 'Night'
END as Time_of_day,
COUNT (DISTINCT order_id) AS Total_orders
FROM `Target_BCS.orders`
GROUP BY Time_of_day
ORDER BY Time_of_day
```

JOB INFORMATION | RESULTS | CHART | JSON

Row	Time_of_day	Total_orders
1	Afternoon	38135
2	Dawn	5242
3	Mornings	27733
4	Night	28331

Insights – Brazilian customers tend to place orders mostly during the afternoon and by a large number (approx. 35% orders more compared to 2nd best time slot). Mornings and night have almost the same quantity and the least during the dawn.

Recommendations –

a) Special offers can be provided to customers during the mornings and dawn to increase the sales during that time.
b) Extra workforce can be deployed during afternoons to help the customers if any help is needed.

- **Evolution of E-commerce orders in the Brazil region:**

1. Get the month-on-month no. of orders placed in each state.

```
SELECT customer_state, EXTRACT (Month from order_purchase_timestamp) AS
Month,
COUNT (order_id) as Total_orders
FROM `Target_BCS.orders` o
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
GROUP BY 1, 2
ORDER BY 1, 2
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTIC

Row	customer_state	Month	Total_orders
1	AC	1	8
2	AC	2	6
3	AC	3	4
4	AC	4	9
5	AC	5	10
6	AC	6	7
7	AC	7	9
8	AC	8	7
9	AC	9	5
10	AC	10	6
11	AC	11	5
12	AC	12	5
13	AL	1	39
14	AL	2	39

Insights – The top 3 states from which maximum orders have come are SP, RJ and MG.

Recommendations –

a) As the brand already has strong presence in these states, the brand should focus on states which have least orders such as RR, AP, AC, AM, RO, TO, SE, AL etc.
b) Promotional campaigns could be run throughout the states, special offers could be run for these specific states to boost the sales and revenue.

2. How are the customers distributed across all the states?

```
SELECT customer_state,
COUNT (customer_unique_id) as Total_cust
FROM `Target_BCS.customers`
GROUP BY customer_state
ORDER BY Total_cust DESC
```

JOB INFORMATION | RESULTS | CHART | JS

Row	customer_state	Total_cust
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020
11	PE	1652

Insights – Maximum customers of Target belong to SP state, RJ and MG come 2nd & 3rd respectively.

Recommendations – If possible, the brand could try opening a few more stores in these states which would help in generating more sales and revenue.

- **Impact on Economy: Analyse the money movement by e-commerce by looking at order prices, freight and others.**

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only). You can use the "payment value" column in the payments table to get the cost of orders.

```
WITH CTE AS

(SELECT
EXTRACT (MONTH FROM order_purchase_timestamp) AS month,
ROUND ((
(
SUM (CASE WHEN EXTRACT (YEAR FROM order_purchase_timestamp) = 2018 THEN
payment_value ELSE 0 END)
-
SUM (CASE WHEN EXTRACT (YEAR FROM order_purchase_timestamp) = 2017 THEN
payment_value ELSE 0 END)
)
/
SUM (CASE WHEN EXTRACT (YEAR FROM order_purchase_timestamp) = 2017 THEN
payment_value ELSE 0 END)
)
* 100, 2) AS Percent_change
FROM `Target_BCS.payments` p
LEFT JOIN `Target_BCS.orders` o
```

```
USING (order_id)
WHERE EXTRACT (MONTH FROM order_purchase_timestamp) BETWEEN 1 AND 8 AND
EXTRACT (YEAR FROM order_purchase_timestamp) IN (2017, 2018)
GROUP BY 1)

SELECT month, CONCAT (Percent_change, '%') as Percent_change
FROM CTE
ORDER BY month
```

JOB INFORMATION | RESULTS | CHART

Row	month	Percent_change
1	1	705.13%
2	2	239.99%
3	3	157.78%
4	4	177.84%
5	5	94.63%
6	6	100.26%
7	7	80.04%
8	8	51.61%

Insights – January 2018 was the month where the sales grew more than 7x when compared to 2017. This could be because Target had recently opened before 2017 so people may not be aware about it and when they were aware the sales grew drastically.

2. Calculate the Total & Average value of order price for each state.

```
SELECT customer_state,
ROUND (SUM (payment_value), 2) AS Total_order_price,
ROUND (AVG (payment_value), 2) AS AVG_order_price
FROM `Target_BCS.payments` p
INNER JOIN `Target_BCS.orders` o
USING (order_id)
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
GROUP BY customer_state
ORDER BY customer_state
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTI

Row	customer_state	Total_order_price	AVG_order_price
1	AC	19680.62	234.29
2	AL	96962.06	227.08
3	AM	27966.93	181.6
4	AP	16262.8	232.33
5	BA	616645.82	170.82
6	CE	279464.03	199.9
7	DF	355141.08	161.13
8	ES	325967.55	154.71
9	GO	350092.31	165.76
10	MA	152523.02	198.86
11	MG	1872257.26	154.71
12	MS	137534.84	186.87
13	MT	187029.29	195.23
14	PA	218295.85	215.92

Insights – SP, RJ & MG have the highest order price but things get interesting when the results are sorted by average order price, they're not even in the top 10 list. PB, AC & RO have the highest average order price. This could mean that customers in SP, RJ & MG are buying more but their average order price is low. This could also mean these states have very high population which contributes to their order price but average buying of customers is less in populated states.

Recommendations – Special offers can be run for bigger populated states so that the average bill value can be increased like providing some extra discounts or some free items on minimum bill value of X amount.

3. Calculate the Total & Average value of order freight for each state

```
SELECT customer_state,
ROUND (SUM (freight_value), 2) AS Total_order_freight_value,
ROUND (AVG (freight_value), 2) AS AVG_order_freight_value
FROM `Target_BCS.order_items` oi
INNER JOIN `Target_BCS.orders` o
USING (order_id)
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
GROUP BY customer_state
ORDER BY customer_state
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTI

Row	customer_state	Total_order_freight_y	AVG_order_freight_v
1	AC	3686.75	40.07
2	AL	15914.59	35.84
3	AM	5478.89	33.21
4	AP	2788.5	34.01
5	BA	100156.68	26.36
6	CE	48351.59	32.71
7	DF	50625.5	21.04
8	ES	49764.6	22.06
9	GO	53114.98	22.77
10	MA	31523.77	38.26
11	MG	270853.46	20.63
12	MS	19144.03	23.37
13	MT	29715.43	28.17
14	PA	38699.3	35.83

Insights - Highest order freight values were spent on SP, RJ & MG respectively. As they have more orders high freight value is justified but they have one of the lowest average freight values in the country. RR, PB & RO have double the average freight value of the states who have the highest freight value. It is very expensive to transport goods in these states compared to other populated bigger states. If somehow this average can be reduced then profit margins can be improved a lot.

Recommendations - Where the average order freight value is higher, discounts & offers can be provided so that the orders increase and if the orders increase the average freight cost will come down when multiple orders are transported together.

- **Analysis based on sales, freight and delivery time.**

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time. Also, calculate the difference (in days) between the estimated & actual delivery date of an order. Do this in a single query.

```
SELECT order_id,
DATE_DIFF (order_delivered_customer_date, order_purchase_timestamp, DAY) AS
time_to_deliver,
DATE_DIFF (order_delivered_customer_date, order_estimated_delivery_date, DAY)
AS diff_estimated_delivery
FROM `Target_BCS.orders`
WHERE order_delivered_customer_date is not null
ORDER BY time_to_deliver
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTIO

Row	order_id	time_to_deliver	diff_estimated_delive
1	e65f1eeee1f52024ad1dcd034...	0	-9
2	bb5a519e352b45b714192a02f...	0	-25
3	434cecee7d1a65fc65358a632...	0	-19
4	d3ca7b82c922817b06e5ca211...	0	-11
5	1d893dd7ca5f77ebf5f59f0d20...	0	-10
6	d5fbeedc85190ba88580d6f82...	0	-7
7	79e324907160caea526fd8b94...	0	-8
8	38c1e3d4ed6a13cd0cf612d4c...	0	-16
9	8339b608be0d84fca9d8da68b...	0	-27
10	f349cdb62f69c3fae5c4d7d3f3...	0	-12
11	f3c6775ba3d2d9fe2826f93b71...	0	-11
12	b70a8d75313560b4acf607739...	0	-9
13	21a8ffca665bc7a1087d31751...	0	-11
14	44558a1547e448b41c48c4087...	1	-5

Insights – For some orders time_to_deliver is 0 which is good but for some other orders it is 209 days. No customer would want to wait so long for their orders to be delivered. The difference between minimum and maximum time for delivery is huge and inconsistent. The data also shows orders have a big difference between estimated delivery date and actual delivery date i.e.- 188 days, which means delivery was late by 188 days and this would affect the reviews given by the customers on the service provided to them. Where as on the other hand few orders were delivered way before estimated delivery date.

Recommendations – The accuracy of the estimated delivery needs to be improved; the data is way off from the actual delivery dates.
The days taken for delivery is huge in some cases, reducing delivery dates can create a good image amongst customers about the service of the brand. Maximum of 7 days should be taken for any delivery. This change in logistics can make customers happy.

2. Find out the top 5 states with the highest & lowest average freight value.

```
(SELECT customer_state,
ROUND (AVG (freight_value), 2) AS AVG_order_freight_value,
'Lowest_5' as Category
FROM `Target_BCS.order_items` oi
INNER JOIN `Target_BCS.orders` o
USING (order_id)
INNER JOIN `Target_BCS.customers` c
```

```
USING (customer_id)
GROUP BY customer_state
ORDER BY AVG_order_freight_value
LIMIT 5)

UNION ALL

(SELECT customer_state,
ROUND (AVG (freight_value), 2) AS AVG_order_freight_value,
'Highest_5' as Category
FROM `Target_BCS.order_items` oi
INNER JOIN `Target_BCS.orders` o
USING (order_id)
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
GROUP BY customer_state
ORDER BY AVG_order_freight_value DESC
LIMIT 5)
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTI

ow	customer_state	AVG_order_freight_v	Category
1	SP	15.15	Lowest_5
2	PR	20.53	Lowest_5
3	MG	20.63	Lowest_5
4	RJ	20.96	Lowest_5
5	DF	21.04	Lowest_5
6	RR	42.98	Highest_5
7	PB	42.72	Highest_5
8	RO	41.07	Highest_5
9	AC	40.07	Highest_5
10	PI	39.15	Highest_5

3. Find out the top 5 states with the highest & lowest average delivery time.

```
(SELECT customer_state,
ROUND (AVG (DATE_DIFF (order_delivered_customer_date,
order_purchase_timestamp, DAY)), 2) AS AVG_time_to_deliver,
'Lowest_5' as Category
FROM `Target_BCS.orders` o
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
WHERE order_delivered_customer_date IS NOT NULL
GROUP BY customer_state
```

```
ORDER BY AVG_time_to_deliver
LIMIT 5)

UNION ALL

(SELECT customer_state,
ROUND (AVG (DATE_DIFF (order_delivered_customer_date,
order_purchase_timestamp, DAY)), 2) AS AVG_time_to_deliver,
'Highest_5' as Category
FROM `Target_BCS.orders` o
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
WHERE order_delivered_customer_date IS NOT NULL
GROUP BY customer_state
ORDER BY AVG_time_to_deliver DESC
LIMIT 5)
```

JOB INFORMATION | RESULTS | CHART | JSON | EXEC

Row	customer_state	AVG_time_to_deliver	Category
1	SP	8.0	Lowest_5
2	MG	12.0	Lowest_5
3	PR	12.0	Lowest_5
4	DF	13.0	Lowest_5
5	SC	14.0	Lowest_5
6	RR	29.0	Highest_5
7	AP	27.0	Highest_5
8	AM	26.0	Highest_5
9	AL	24.0	Highest_5
10	PA	23.0	Highest_5

Insights – When highest & lowest average delivery time is compared, it is very evident that the difference between them is more than 3x. In some states it takes 8-12 days to deliver good whereas in others 27-29 days. The waiting time for customers is huge and it should be cut short.

Recommendations – Logistics need to be improved to bring the average delivery time below 7 days in today's well-connected world. Faster delivery time = good service. Instead of delivery by road, rail network can be preferred and for longer distances air transport can be used which will reduce delivery time drastically.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```
SELECT customer_state,
ROUND (AVG (DATE_DIFF (order_estimated_delivery_date,
order_delivered_customer_date, DAY))) AS Diff_delivery_estimated_days
FROM `Target_BCS.orders` o
INNER JOIN `Target_BCS.customers` c
USING (customer_id)
WHERE order_status = 'delivered'
GROUP BY customer_state
ORDER BY Diff_delivery_estimated_days DESC, customer_state
LIMIT 5
```

JOB INFORMATION | RESULTS | CHART | JS

ow	customer_state	Diff_delivery_estimat
1	AC	20.0
2	AM	19.0
3	AP	19.0
4	RO	19.0
5	RR	16.0

Insights – These are the top 5 states where orders get delivered very fast as compared to estimated delivery time and the average difference is huge which means the parameter to measure to estimated delivery time is not accurate at all. Showing 30+ days for estimated delivery is not a good sign of the best service provided by the brand.

Recommendations – Work should be put in to check why the estimated date is so far and different from actual delivery date. What parameters are considered for calculation of the date need to reconsidered. The data accuracy has to be improved.

- **Analysis based on the payments:**

1. Find the month-on-month no. of orders placed using different payment types.

```
SELECT payment_type,
EXTRACT (Month from order_purchase_timestamp) AS Month,
COUNT (order_id) as Num_of_orders_placed
FROM `Target_BCS.orders` o
INNER JOIN `Target_BCS.payments` p
USING (order_id)
GROUP BY payment_type, Month
ORDER BY payment_type, Month
```

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTI

ow	payment_type	Month	Num_of_orders_plac
1	UPI	1	1715
2	UPI	2	1723
3	UPI	3	1942
4	UPI	4	1783
5	UPI	5	2035
6	UPI	6	1807
7	UPI	7	2074
8	UPI	8	2077
9	UPI	9	903
10	UPI	10	1056
11	UPI	11	1509
12	UPI	12	1160
13	credit_card	1	6103
14	credit_card	2	6609

Insights – Credit card is the most preferred mode of payment for customers in Brazil. UPI comes in at 2nd position.

Recommendations – Credit card company representative booths can be setup inside the store so that people can be made aware of the credit card offers to use them more and more. Also, customers who don't own a credit card can be convinced to buy them and use them for shopping.

2. Find the no. of orders placed on the basis of the payment instalments that have been paid.

```
SELECT payment_installments,
```

```
COUNT (order_id) as Total_orders,
FROM `Target_BCS.payments`
WHERE payment_installments != 0
GROUP BY payment_installments
ORDER BY payment_installments
```

JOB INFORMATION RESULTS CHAI

Row	payment_installment	Total_orders
1	1	52546
2	2	12413
3	3	10461
4	4	7098
5	5	5239
6	6	3920
7	7	1626
8	8	4268
9	9	644
10	10	5328
11	11	23
12	12	133
13	13	16
14	14	15

Insights – Most of the orders have 1 payment instalment completed. The highest payment instalment is 24 for 18 orders. This could mean that most customers don't prefer large number of instalments.